

ONE MIDPOINT ENGINE • THREE PROGRESSIONS • MACHINE-VERIFIED

AP

GP

HP

The Omni Series

AP • GP • HP

Arithmetic, geometric and harmonic power-sums under a single midpoint moment-block expansion — one derivation, one correction library, three specializations distinguished only by what closes.

Formulas with Space Exploration • Data Storytelling

One expansion, recentered on the midpoint

Recenter any sum on $M = (F+L)/2$ and expand in symmetric offsets. Odd terms cancel; what survives are the moment blocks $\mu_{2m}(X)$. This one identity is the whole engine:

$$\sum f(k) = X \cdot \sum_m \left[f^{(2m)}(M) / (2m)! \right] \cdot \mu_{2m}(X)$$

$k = F \dots L$ · the same $\mu_{2m}(X)$ library serves all three families

AP

$$f(k) = k^d$$

terminates → **exact polynomial**

The Pascal binomial $C(d, 2m)$ is finite. This is Faulhaber, in factored midpoint form.

GP

$$f(k) = (a \cdot r^k)^s$$

native geometric closed form

Closes as $a^s(r^{s^n}-1)/(r^s-1)$; also fits the lens with $R = r^s$, and converges.

HP

$$f(k) = k^{-s}$$

asymptotic → **ζ / ψ exact**

The generalized binomial $(s)_{2m}/(2m)!$ never terminates — exact via zeta/polygamma.

New reframing — not new theorems, and honest about it

The classical picture

Faulhaber (1631)

sums of integer powers — a separate polynomial per power

Euler–Maclaurin

a general correction series, treated case by case

Hurwitz ζ / polygamma

the exact home of reciprocal-power sums, on its own

The Omni reframing

One shared μ -library

the AP moment blocks are reused verbatim inside the HP expansion

One derivation

the same five-step midpoint symmetry generates all three families

Verified + constant-time

packaged as an $O(1)$ primitive with exhaustive machine checks

Honest positioning: the quantities are classical. The contribution is the unification, the shared correction library, and the verified constant-time packaging — not a claim that harmonic sums have a finite closed form. They provably do not.

The three families, from formula to number

AP

GENERIC

$$D^d = X \cdot \sum C(d, 2m) M^{d-2m} \mu_{2m}(X)$$

SPECIFIC

$$d = 3, k = 3 \dots 7 \quad (X=5, M=5)$$

$$5 \cdot 5 \cdot [5^2 + 3\mu_2] = 5 \cdot 5 \cdot 31$$

$$= 775$$

GP

GENERIC

$$GP[s] = a^s (r^{sn} - 1) / (r^s - 1)$$

SPECIFIC

$$s = 3, a=1, r=2, n=5$$

$$1^3 + 2^3 + 4^3 + 8^3 + 16^3$$

$$= 4,681$$

HP

GENERIC

$$HP[s] = \zeta(s, F) - \zeta(s, L+1)$$

SPECIFIC

$$s = 3, k = 3 \dots 7$$

$$3^{-3} + 4^{-3} + 5^{-3} + 6^{-3} + 7^{-3}$$

$$= 0.06820712$$

Explicit midpoint formulas, and their values over $k = 3 \dots 7$

D^1	$X \cdot M$	25
D^2	$X \cdot [M^2 + \mu_2]$	135
D^3	$X \cdot M \cdot [M^2 + 3\mu_2]$	775
D^4	$X \cdot [M^4 + 6M^2\mu_2 + \mu_4]$	4,659
D^5	$X \cdot M \cdot [M^4 + 10M^2\mu_2 + 5\mu_4]$	28,975
D^6	$X \cdot [M^6 + 15M^4\mu_2 + 15M^2\mu_4 + \mu_6]$	184,755
D^7	$X \cdot M \cdot [M^6 + 21M^4\mu_2 + 35M^2\mu_4 + 7\mu_6]$	1,200,175
D^8	$X \cdot [M^8 + 28M^6\mu_2 + 70M^4\mu_4 + 28M^2\mu_6 + \mu_8]$	7,907,139
D^9	$X \cdot M \cdot [M^8 + 36M^6\mu_2 + 126M^4\mu_4 + 84M^2\mu_6 + 9\mu_8]$	52,666,255
D^{10}	$X \cdot [M^{10} + 45M^8\mu_2 + 210M^6\mu_4 + 210M^4\mu_6 + 45M^2\mu_8 + \mu_{10}]$	353,814,675

Coefficients are Pascal's row d at even positions; the series terminates — exact integers at every power.

One geometric closed form, evaluated for s = 1...10

$GP[s] = a^s(r^{sn}-1)/(r^s-1)$ worked with a=1, r=2, n=5 → $1 + 2^s + 4^s + 8^s + 16^s$

s = 1	31	s = 2	341
s = 3	4,681	s = 4	69,905
s = 5	1,082,401	s = 6	17,043,521
s = 7	270,549,121	s = 8	4,311,810,305
s = 9	68,853,957,121	s = 10	1,100,586,419,201

The power s enters only through r^s; the sum stays an exact finite geometric series at every power — the simplest of the three to close.

The polygamma form, and its values over k = 3...7

$$\text{HP}[s] = (-1)^s/(s-1)! \cdot [\psi^{(s-1)}(F) - \psi^{(s-1)}(L+1)] \quad (s = 1 : \psi(L+1) - \psi(F))$$

s=1	$\psi(L+1) - \psi(F)$	1.09285714	s=2	$+ [\psi^{(1)}]$	0.26179705
s=3	$- \frac{1}{2} [\psi^{(2)}]$	0.06820712	s=4	$+ \frac{1}{6} [\psi^{(3)}]$	0.01904003
s=5	$- \frac{1}{24} [\psi^{(4)}]$	0.00559989	s=6	$+ \frac{1}{120} [\psi^{(5)}]$	0.00170982
s=7	$- \frac{1}{720} [\psi^{(6)}]$	0.00053587	s=8	$+ \frac{1}{5040} [\psi^{(7)}]$	0.00017100
s=9	$- \frac{1}{40320} [\psi^{(8)}]$	0.00005526	s=10	$+ \frac{1}{362880} [\psi^{(9)}]$	0.00001801

No power collapses to a finite algebraic form — but each is an exact special-function value, matched to 40+ digits.

Constant cost, at any size or power

$O(1)$

per evaluation

a fixed handful of operations, independent
of the number of terms or the power

10^{30}

terms, no cost growth

brute force $\approx$ 32 trillion years; the formula
answers in under a millisecond

25+

exact digits

AP integer totals past 25 digits (D^{30}); HP
 $\sim 10^{-15}$ with six correction blocks

Where each family lands

AP $\rightarrow$ exact integers, unbounded size. **GP** $\rightarrow$ exact, one geometric step. **HP** $\rightarrow$ exact via ζ/ψ , or asymptotic to machine precision in constant time.

Answer size grows ~ 4 digits per added lattice dimension — past standard floating point beyond ~ 5 dimensions, where exact arithmetic is the point.

Test cases run — zero failures

97,200

AP tests (prior work)

every (F,L) with $F \leq L \leq 80$ across powers
 $D^1 \dots D^{30}$

36,600

HP tests (this work)

every (F,L) with $F \leq L \leq 60$ across powers s
 $= 1 \dots 20$

0

failures, combined

closed form vs direct summation — exact
in every single case

133,800+ exhaustive cases across both families, 0 failures. Half-integer midpoints (even X) and an independent polygamma cross-check included; at 80-digit arithmetic the worst discrepancy is 10^{-54} — pure evaluation noise, not a formula error.

Promising directions — grounded, not grandiose

1

Wider families

Alternating series, arbitrary common-difference progressions, and mixed powers under the same midpoint engine.

2

Auto-generated library

Symbolic Bernoulli recurrence emits the μ -blocks to any depth on demand — no hand tuning, no ceiling.

3

Convergence acceleration

Pairing the asymptotic HP tails with resummation to push accuracy further at small denominators.

4

Lattice-scale computing

A constant-time exact-aggregation layer for cosmological grids, octree debris indexing, and error-correcting code sizing.

The promise is a single, verified, constant-time primitive for counting over structured sequences — useful exactly where a lattice must be aggregated, and honest about where arithmetic stops.

THE OMNI SERIES

One midpoint engine. Three progressions. Zero failures.

AP exact · GP exact · HP exact where exact, constant-time where not.

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